



An oscilloscope is an indispensable tool for any engineer who is working on design, test or even repair of any electronics equipment. It helps to visualise the functioning of an electronics circuit and can act as an engineer's eye to the electronics world. Therefore it is important to carefully choose the right oscilloscope, which would be required to capture the signal of interest” — Akash Srivastava, application engineer, Keysight Technologies India Pvt Ltd



Apart from technical features, a unique aspect that some oscilloscope makers are incorporating is assistance for students and professors. For instance, Keysight Technologies' DSO1000A/B series provides training tools such as education student lab guides and tutorials, and professor slide sets of oscilloscope fundamentals.

Tektronix's TBS1000B-EDU series come with an integrated courseware

TABLE II
Some USB/PC Based Oscilloscopes

Company	Product	Channels	Bandwidth	Sampling rate	Display memory depth	Price
Bitscope	Bitscope Micro (oscilloscope + logic analyser)	Two analogue, six digital	20MHz	40MSa/S	—	₹ 11,641*
Pico Technology	PICOSCOPE 2204A-D2	Two analogue	10MHz	100MSa/S	8kpts	₹ 15,312*
Pico Technology	PICOSCOPE 2205A-D2	Two analogue	25MHz	200MSa/S	16kpts	₹ 19,502*
Hantek	DSO3064	Four analogue	60MHz	200MSa/S	10kpts-16Mpts	₹ 27,490*
Tenma	72-10155	Two analogue	40MHz	100MSa/S	64kpts	₹ 29,936*
MEILHAUS	DS1M12	Two analogue	250kHz	20MSa/S	32kpts	₹ 39,933*

*Price listed as on e-commerce websites like element14.com and company websites

TABLE III
Interesting Miniature, Pocket-Size Oscilloscopes

	Oscium's iMSO-204 and iMSO-204L (lightning-compatible) are mixed-signal oscilloscopes designed specifically for the iPhone, iPad and iPod. Highlights: Two analogue + four digital channels, sample rate: 50MSa/S, bandwidth: 5MHz, 200ns/div-10s/div Price: US\$ 399.97
	Gobotronics' XminiLab Portable is a small mixed-signal oscilloscope with an arbitrary waveform generator and protocol sniffer. Highlights: Two analogue inputs, maximum sampling rate: 2MSa/S, analogue bandwidth: 200kHz, resolution: 8-bits, input impedance: 1MΩ 15pF, buffer size per channel: 256, input voltage range: -14V to +20V Price: US\$ 118
	LabNation's SmartScope is a 100MSa/s open source oscilloscope for iPad, Android and PC. A must-have for every Arduino and Raspberry Pi developer. Highlights: 2x100MSa/s 45MHz oscilloscope, 50MSa/s arbitrary waveform generator digital-logic analyser at 100MSa/s, digital waveform generator at 100MSa/s, 200 waveforms/second data updates Price: US\$ 229
	Gobotronics' oscilloscope watch Highlights: Two analogue inputs, maximum sampling rate: 4MSa/s, analogue bandwidth: 200kHz, resolution: 8-bits Price: US\$ 150
	RedPitaya Highlights: DIY oscilloscope with dual channels with a sampling rate of 125MSa/s, spectrum analyser with 50MHz, 14-bit arbitrary waveform generator, frequency response analyser with 60MHz bandwidth and a 2x2 MIMO PID programmable controller Price: ₹ 46,750

*Price listed as on e-commerce websites like element14.com and company websites